Alexander Bainbridge

Technical Lead | Cloud-Native Architect | AI-Driven Engineer

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🚀 About Me

**Technical Lead	Cloud-Native Architect	AI-Driven Engineer**

I'm a technical lead and polyglot engineer with **10 years' experience** building **cloud-native platforms**, leading **high-performing teams**, and delivering **AI-driven innovation**. Passionate about solving hard problems, mentoring engineers, and architecting scalable, future-ready systems.

🛠️ Tech Stack

PYTHON | .NET | TYPESCRIPT | REACT | AZURE | AWS | TERRAFORM | DOCKER

💼 Professional Experience

Duel — *Senior Full-stack Engineer*

Oct 2025 – Feb 2026 | Bristol, Hybrid

- Built out an AI first advocate challenge creation and content creation analysis prototype using langraph
- Productionised the content creation analysis in their unfamiliar stack. Released to several brands
- Held a retro in the first month focussing on psycological safety

- Assisted in BAU such as adding in new translations in short time frames for brand launches
- Disseminated a years worth of AI enabled engineering learnings to the wider engineering team
- Built out AI-enabled engineering - rules, skills and workflows

Clinithink — *Technical Lead*

Jun 2023 – Aug 2025 | Bridgend, Hybrid

- Led a small team to architected and deliver **CustomerHub** in a quarter (ahead of schedule), re-platforming a legacy .NET/Azure app into **HubSpot CMS + React + Azure Functions**, **cutting deployment cycles by 75%**. .Net was new to me
- Built **BuildTools** — a PowerShell cmdlet suite and reusable **Azure DevOps pipeline templates** to standardise pipelines across the business. Provided necessary tooling for across the business, such as tracking our AI coding metrics, generating change logs, versioning, auto promotion of artifacts in **Azure Devops**
- Led **ProjectCipher**, an AI-driven experiment using **ML.NET + TypeScript + React** to evaluate one-shot full-stack generation.
- Mentored engineers and drove **psychological safety**, boosting delivery velocity and collaboration.
- Was a member of the Wellbeing team to promote openness and inclusivity
- Performed multiple demo’s within our technical forum across a number of different projects and experiments
- A brief stay within the **CLiX Unlock** unlock teams, two crossfunctional teams who reported to a colleague as tech lead. I naturally re-positioned myself into a senior engineer role to support the lead.

Clinithink — *Software Engineer*

Jun 2021 – Jun 2023 | Bridgend, Hybrid

- **CLiX Revenue**, an already established product **Python/Django/React** app surfacing NLP-powered insights. Django was new to me. Eventually led this team to became the **engineering benchmark** at Clinithink.
- Created Clinithink’s **first CI/CD pipeline** using **Terraform + Azure DevOps**, **cutting QA cycle times from days to hours**.
- Partnered with product owners and scrum masters to define features, improve agile processes, and coach peers on modern engineering practices.

Raytheon — *Senior Software Engineer*

Nov 2020 – Apr 2021 | Cheltenham

- Built an **employee portal** using **React, Node.js, MySQL, Liquibase**, and **Docker**, deployed via **Jenkins + Kubernetes + Helm**.
- Improved test coverage and quality using **Jest, Cypress, and Cucumber**.
- Mentored junior engineers and influenced DevOps adoption across teams.

BAE Systems Applied Intelligence — *Software Engineer*

Aug 2016 – Nov 2020 | London & Cheltenham

- Built a **distributed ETL framework** in **Python**, integrating **Kafka, Redis, Elasticsearch, and NiFi** to automate data processing.
- Designed CI/CD pipelines via **Ansible** and delivered robust testing suites using **Pytest**.
- Provided 24/7 support for critical client systems, resolving high-severity incidents under strict SLAs.

🌟 Highlight Projects

Project	Tech	Summary
ProjectVelour	FastAPI • ComfyUI • Stable Diffusion • React	AI-driven platform for personal expression through tattoos and nail art.
Spelltail	Unity • C# • LLMs	A **Snake x Scrabble** mashup with AI-driven story generation. Devlog
ProjectCipher	ML.NET • TypeScript • React	Experimental AI-assisted **full-stack code generation**. ProjectCipher
BuildTools	PowerShell • Azure DevOps • Python	A cmdlet + pipeline toolkit that standardised devops practices.

📜 Certifications

- Mental Health First Aider

🎓 Education

University of South Wales — BSc (Hons) Computer Games Development, 2:1 *(2012–2016)*
Pontypool College — HND Graphic Design *(2005–2008)*

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